

Hardware Reproducibility Noise Decays with Training: Measuring Backend Effects Early Overstates Them by an Order of Magnitude

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Retraining an identical model with an identical seed on a different compute backend yields a different validation curve, and it is often assumed that this irreproducibility is of the same order as the seed-to-seed variation that papers already report. We measure it directly. Eight seeds of a linear-attention model were trained to 7,500 steps on a Tesla T4 GPU and retrained on CPU with identical code, data, and seeds, yielding 120 paired comparisons. The mean absolute backend difference falls from 0.0331 over steps 500–2,500 to 0.0060 over steps 3,000–7,500, a $5.5\times$ decay: trajectories start apart and converge. Relative to the between-seed standard deviation *at the same step*, the backend term is $1.30\times$ in the transient but only $0.53\times$ in the late regime, where architecture decisions are actually taken. No directional bias is detectable ($t(7) = -1.42$, $p = .198$; 64 of 120 signed differences negative). The practical consequence is a measurement caveat rather than a reproducibility alarm: because between-seed noise is itself non-stationary—its standard deviation falls from 0.101 at step 500 to 0.011 at step 7,500, a factor of nine—any backend comparison anchored to a single global noise estimate is off by a large factor. We first made that error ourselves and reported an inflated ratio of $3.00\times$; the correction is documented here because the failure mode is easy to reproduce and hard to notice.

Keywords: reproducibility, floating-point non-determinism, hardware variability, early stopping, experimental methodology

The question

Two laboratories run the same code, with the same seed, on different GPUs, and obtain different numbers. This is uncontroversial: floating-point reductions are

not associative, and kernel selection differs across devices. What is not established is *how large* the resulting disagreement is relative to the variation that papers already quantify, namely the spread across random seeds. If the backend term were comparable to or larger than the seed term, then the customary “mean $\pm$ SD over N seeds” would omit a term of the same order, and architecture comparisons whose reported margins are of that size would not be reproducible across machines.

We tested this on a concrete system rather than in general, and the answer depends critically on *when* the measurement is taken.

Method

The subject is condition delta of a pre-registered capacity experiment: a small linear-attention model (193,493 parameters) trained on a multi-query associative recall task, with validation accuracy

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evaluated every 500 steps at sequence loads 96 and 128. Eight seeds had already been trained to 10,000 steps on a Tesla T4 GPU under JAX, with `jax_default_matmul_precision=highest`, `-xla_gpu_deterministic_ops=true` and `TF_CUDNN_DETERMINISTIC=1`.

Each seed was then retrained from scratch on CPU (Intel i3-7100, four threads) with identical code, identical data generation, and the identical seed, to 7,500 steps. This yields 15 paired evaluation points per seed and $8 \times 15 = 120$ paired comparisons. The comparison is *within seed*: every CPU value is compared against the GPU value of the same seed at the same step, so the seed contribution is differenced out by construction.

Two quantities are reported at each step s : the mean absolute backend difference $\overline{|d_s|} = \frac{1}{8} \sum_i |a_{i,s}^{\text{CPU}} - a_{i,s}^{\text{GPU}}|$, and the between-seed standard deviation of the GPU runs *at that same step*, σ_s . Their ratio answers the question of interest: how much noise does changing the backend add, in units of the noise the field already reports?

Results

The backend effect decays with training

Table 1

Backend difference and between-seed dispersion, by training step

Step	$\overline{ d_s }$	σ_s	Ratio
500	0.0353	0.0972	0.36
1,000	0.0362	0.0702	0.52
1,500	0.0442	0.0371	1.19
2,000	0.0275	0.0092	2.98
2,500	0.0224	0.0133	1.68
3,000	0.0199	0.0112	1.78
3,500	0.0070	0.0145	0.48
4,000	0.0060	0.0188	0.32
5,000	0.0049	0.0090	0.54
6,000	0.0030	0.0118	0.25
7,000	0.0039	0.0103	0.38
7,500	0.0035	0.0107	0.32

Note. $\overline{|d_s|}$ is the mean absolute CPU–GPU difference in validation accuracy across eight seeds; σ_s is the between-seed standard deviation of the GPU runs at the same step. Rows for 4,500, 5,500 and 6,500 omitted for space; the full table is in the repository.

Grouping into regimes, the mean absolute difference is 0.0331 over steps 500–2,500 and 0.0060 over steps 3,000–7,500, a decay of 5.5 $\times$. Expressed relative to between-seed dispersion at the same step, the backend term averages 1.30 $\times$ in the transient and 0.53 $\times$ in the late regime. Over all 120 comparisons the ratio is 0.81 $\times$.

The interpretation is that the two trajectories are not diverging but converging. Different arithmetic on different devices separates the runs early, when the loss surface is being traversed quickly and small perturbations are amplified; as training settles, both runs approach the same region and the difference shrinks.

No directional bias

Of the 120 signed differences, 64 are negative. Averaging within seed and testing the eight per-seed means against zero gives $t(7) = -1.42$, $p = .198$. There is no evidence that CPU systematically underperforms or outperforms the GPU; what the backend contributes is dispersion, not a shift. This conclusion was stable across sample sizes ($n = 3, 4, 8$) and across both horizons, and it is the most robust finding of the experiment.

A control that came for free

One seed lacked a checkpoint and was retrained from zero on the same machine. It reproduced 0.5182 at step 500 against 0.51822 in the original CPU run. Within a backend, training is exactly reproducible; the differences measured here therefore arise from the change of device and not from run-to-run variability.

The error we made first, and why it is worth reporting

Our first analysis compared the backend differences, which were measured over steps 500–2,500, against a *single* between-seed standard deviation of 0.01105—the value at step 7,500. That produced a ratio of 3.00 $\times$ and a worst case of 20.25 $\times$, and supported the conclusion that backend choice dominates seed choice.

It is wrong, because between-seed dispersion is not a property of the experiment but a function of the step:

Step	500	1,000	1,500	2,000	2,500	5,000	7,500
σ_s	.101	.073	.038	.010	.014	.009	.011

A factor of nine between the transient and the late regime. Anchoring an early-regime numerator to a late-regime denominator inflates the ratio; pairing each step against its own σ_s brings $3.00\times$ down to $1.30\times$ over the same steps. The reported worst case was the same artifact: the largest single difference, 0.224, occurs at step 1,500 of one seed, where $\sigma_s = 0.038$, not 0.011.

We report this because the failure mode is easy to reproduce, invisible in the output, and would have led us to publish the opposite of what the extended measurement shows. Any study that summarizes non-stationary noise with one number is exposed to it.

Limitations

The result is specific to one architecture, one task, one GPU model and one CPU. It does not separate “CPU versus GPU” from “this CPU versus this GPU,” and it says nothing about mixed-precision training, distributed reductions, or larger models, where the mechanism may differ in magnitude or in sign. The CPU runs were executed with the processor frequency capped at 75% for thermal reasons; this changes wall-clock time per step, not arithmetic. Finally, $p = .198$ with eight seeds does not establish the absence of bias—it establishes that any bias is small relative to dispersion.

Implications

The headline is a measurement caveat, not a reproducibility alarm. In the regime where architecture com-

parisons are actually decided, changing backend contributed roughly half the variation of changing seed, and about 30% of the decision margin of the study that hosted this experiment. Reporting “mean $\pm$ SD over N seeds” is not meaningfully incomplete on that account.

The transferable point is narrower and, we think, more useful: *reproducibility studies must declare the training regime in which they measured*. A backend comparison conducted at 2,500 steps and one conducted at 7,500 steps on the same system differ by a factor of five, and the early number is the alarming one. Absent that declaration, a reader cannot tell whether a reported irreproducibility is a property of the system or an artifact of when the ruler was applied.

Open Practices

The training code, the stored validation histories for all sixteen runs, the analysis script that regenerates every number in this note, and the full step-by-step table are available at github.com/SperanzaMax/telar-ligamento under `resultados/backend/`. The non-stationarity of validation noise documented in Section 5 motivated a feature of the `stoppower` package (<https://doi.org/10.5281/zenodo.21711767>), which requires the user to declare the step range over which noise is estimated and warns when the constant-noise assumption fails within a window. The pre-registered study that supplied the runs is reported separately (<https://doi.org/10.5281/zenodo.21630279>).